Recurrence Diagnostics for Econometric Stress Episodes: A Methods Paper with a Cluster-Adjusted Stress Functional

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Abstract

Standard stress diagnostics in applied econometrics typically measure the magnitude of variation, tail losses, or event frequency, but not the temporal organization of extreme episodes. That omission matters whenever policy or risk interpretation depends on whether extremes arrive as isolated shocks or as clustered stress. This paper develops a reproducible recurrence-oriented framework for stress diagnostics and introduces a cluster-adjusted stress functional that combines exceedance probability with a finite-threshold clustering coefficient. The paper has two goals. First, it provides a serious methods contribution: a formal probabilistic setup, explicit estimators for exceedance incidence, return-time behavior, runs-based clustering, and multivariate stress activation, together with a reproducible computational implementation. Second, it adds a narrow theoretical contribution: basic ordering and plug-in consistency results for the proposed cluster-adjusted functional under standard high-threshold regularity conditions, together with an asymptotic motivation drawn from clustered exceedance limits. The synthetic evidence reported below shows that recurrence diagnostics separate benchmark processes that would be inadequately summarized by volatility alone. In the current artifact set, the estimated finite-level clustering coefficients for two stress benchmarks are 0.673 and 0.780, the logistic benchmark observable has normalized return-time coefficient of variation 0.795, the clustered synthetic benchmark has normalized return-time coefficient of variation 1.286, and rolling volatility is elevated during exceedance periods without exhausting the clustering information. The paper makes modest claims: it does not offer new asymptotic theory for extremal-index estimation itself, nor does it claim structural macroeconomic identification. Its contribution is a mathematically careful and computationally reproducible bridge between extreme-value diagnostics and econometric stress analysis.

Keywords: extreme values, extremal index, recurrence, return times, stress testing, time series econometrics

1 Introduction

The motivating problem is straightforward. In many macro-financial and econometric applications, a "stress episode" is treated as a period in which a variable becomes large in magnitude, volatile, or both. But a stress episode is not defined only by how large observations become. It is also defined by how they recur. A series that produces isolated tail observations conveys a different kind of fragility from a series that produces clustered extremes, even if the two series have comparable event counts or similar rolling volatility.

That distinction is conceptually obvious and methodologically underused. Standard variance-based diagnostics, from ARCH and GARCH models onward, are designed to summarize time-varying second moments [8, 4]. Tail-risk and systemic-risk measures such as VaR, CoVaR, expected shortfall, and capital-shortfall metrics focus on the severity of bad states and their cross-sectional comovement [1, 2, 5, 3].

Yet the temporal spacing of threshold exceedances is a different object. Extreme-value theory for dependent sequences treats exceedance clustering and return times as first-order features of the data-generating process rather than as narrative afterthoughts [13, 7, 6]. The extremal index provides a canonical summary of tail clustering [15, 9, 16]. Related recurrence ideas also appear in the study of hitting times and rare events for stochastic and dynamical systems [11, 10, 14]. What is comparatively rare in applied econometrics is a workflow that makes these objects operational in a transparent, reproducible way and places them next to familiar volatility baselines rather than outside the analysis.

The gap addressed here is therefore narrower, and in some ways more demanding, than a generic shortage of methods. Extreme-value theory already provides a mature language for thinking about exceedance clustering, return times, and dependence in the tail, while econometric practice already offers highly developed tools for modeling volatility, downside risk, and financial stress. What remains comparatively underdeveloped is a framework in which recurrence itself is treated as an object of stress measurement and is analyzed alongside, rather than outside, standard econometric summaries. That is the problem taken up in this paper. I formalize a recurrence-based stress framework built from exceedance incidence, return-time behavior, runs-based clustering, multivariate stress activation, and volatility baselines; I implement those objects in a transparent computational pipeline that records intermediate artifacts rather than only final graphics; and I introduce a cluster-adjusted stress functional that distinguishes between stress processes with similar incidence but different temporal concentration of extremes. The theoretical contribution is intentionally narrow, but it is not decorative: the functional is given a careful finite-threshold interpretation, an asymptotic motivation, basic ordering properties, and plug-in consistency under standard regularity conditions. The broader aim is not to compete with the deep asymptotic theory of dependent extremes, but to make part of that theory usable in the setting of applied stress diagnosis.

The paper's empirical claims remain modest. The current evidence consists of synthetic experiments and a small multivariate illustration, not a structural macroeconomic application. The scientific claim is therefore limited to the one the evidence can bear: recurrence diagnostics recover dimensions of stress persistence and organization that are not exhausted by volatility summaries alone.

The remainder of the paper is organized as follows. Section 2 reviews the relevant literature and identifies the specific gap addressed here. Section 3 introduces the probabilistic framework and the estimands. Section 4 describes the estimators and the implementation logic. Section 5 develops the cluster-adjusted stress functional and states basic theoretical results. Section 6 presents synthetic evidence from the generated artifacts. Section 7 discusses limitations and scope.

2 Literature Review

2.1 Extreme values under dependence

The first strand is the theory of extremes for dependent sequences. Classical presentations of EVT often begin with independent observations because independence exposes the basic tail mechanisms cleanly, but that simplification is rarely adequate for economic time series, where temporal dependence is often strongest precisely in turbulent periods. Leadbetter, Lindgren, and Rootzén [13] provided the foundational framework for extremes under dependence, including the conditions under which clustering enters the limiting behavior of maxima and exceedances. Embrechts, Klüppelberg, and Mikosch [7] and Coles [6] extended that framework into the language of applied risk analysis, making clear that one cannot infer the meaning of rare events from marginal tails alone. For the present paper, the lesson is specific: once dependence is admitted, the number of threshold exceedances no longer exhausts what must be said about stress. Their temporal arrangement becomes part of the estimand.

2.2 Extremal index and clustering diagnostics

The second strand concerns the extremal index and the practical problem of quantifying clustering. O'Brien [15] established the extremal index as a scalar summary of local tail dependence for stationary sequences, thereby giving formal expression to the distinction between isolated and bunched extremes. Later work by Ferro and Segers [9] and Süveges [16] developed estimation strategies that made the object empirically usable, while also underscoring the fragility induced by threshold choice, finite samples, and declustering rules. This literature is indispensable for the present paper, although I do not attempt to improve it on its own terms. My interest lies instead in a more applied question: how should clustering diagnostics be incorporated into stress analysis so that they do not remain isolated technical appendices to otherwise conventional volatility studies? The answer pursued here is to treat runs-based clustering not as a side calculation, but as one ingredient of a broader recurrence-based description of stress.

2.3 Recurrence and return-time analysis

The third strand studies recurrence itself, especially through hitting times and return-time distributions. In dynamical-systems and stochastic-process settings, these quantities have long been understood as structurally meaningful rather than merely descriptive. Hirata, Saussol, and Vaienti [11] showed how return-time statistics encode recurrence properties in a general framework, while Freitas, Freitas, and Todd [10] and Lucarini et al. [14] deepened the connection between rare events, recurrence, and temporal dependence. What matters for the present argument is not the full technical range of that literature, but its conceptual insistence that the spacing of rare events carries information distinct from their marginal severity. Applied econometrics has only partially absorbed that point. Return times are often discussed informally in crisis narratives, yet they are much less often estimated and reported as formal objects inside empirical stress workflows.

2.4 Volatility econometrics and tail-risk baselines

The fourth strand is the econometric baseline against which the present framework should be understood. ARCH and GARCH models [8, 4] established time-varying volatility as a central empirical object, and they remain indispensable whenever the aim is to characterize changing second-moment conditions. Tail-risk measures and conditional downside summaries, including CoVaR [1], extend that baseline toward questions of severity and dependence in bad states. None of this is in dispute. The issue is simply that these tools answer a different question from the one pursued here. They are designed to measure scale, exposure, and tail losses, whereas recurrence diagnostics are designed to measure the temporal organization of repeated exceedances. A period of elevated volatility may contain a single shock, a sequence of loosely related shocks, or a tightly clustered stress episode. Treating those cases as equivalent is often a stronger modeling choice than the econometric discussion acknowledges.

2.5 Macro-financial stress monitoring and systemic-risk measurement

The fifth strand is the literature on stress monitoring and systemic risk. Surveys such as Bisias et al. [3] make clear that this literature is already rich in market-based, balance-sheet-based, and network-oriented indicators. Measures such as CoVaR [1], SRISK [5], and systemic expected shortfall [2] are designed to quantify the severity of distress, the capital shortfall associated with it, and the extent to which risk propagates across institutions. Those are major achievements, and the present paper is not offered as a competitor to them. It should instead be read as targeting a narrower diagnostic dimension that those measures do not make central, namely the recurrence profile of stress through time. The question here is not how much capital is at risk in a bad state, but how stress episodes reappear, bunch together, and persist once threshold-based distress has begun.

2.6 Gap and contribution

Taken together, these literatures leave a surprisingly specific opening. The missing piece is not theory about dependent extremes, nor technique for volatility or systemic-risk measurement. It is an analytic framework in which recurrence, clustering, and return-time structure are treated as explicit stress objects and are estimated within the same reproducible workflow as conventional econometric summaries. That is the contribution pursued in this paper. I formulate a recurrence-based stress framework in probabilistic terms, implement it through a deterministic computational pipeline, and use it to motivate a cluster-adjusted stress functional that preserves incidence information while penalizing temporal concentration of extremes. The result is not a new general theory of stress episodes, but a more coherent way to connect established theory on dependent extremes to the practical language of econometric stress analysis.

3 Probabilistic Framework

Let $\{X_t\}_{t\in\mathbb{Z}}$ be a strictly stationary real-valued process on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ with marginal distribution function F. The object X_t may represent a transformed return, a stress indicator, or another scalar measure whose upper tail corresponds to more severe conditions. The multivariate extension is introduced below.

For a threshold u, define the exceedance indicator

$$I_t(u) = \mathbf{1}\{X_t > u\}.$$

The exceedance probability is

$$p(u) = \mathbb{P}(X_0 > u).$$

In finite samples of length n, high-threshold analysis for estimation should work with an intermediate threshold sequence u_n satisfying

$$k_n := n\,\overline{F}(u_n) \to \infty, \qquad \frac{k_n}{n} \to 0,$$

where $\overline{F}(u) = 1 - F(u)$. This keeps exceedances rare but observable while still allowing their count to diverge. By contrast, the bounded-count regime

$$n\,\overline{F}(u_n) \to \tau \in (0, \infty)$$

belongs to point-process limit theory for rare events and is not the estimation regime used for the consistency claim below.

If $t_1 < \cdots < t_{N_n}$ denote the exceedance times among $\{1, \ldots, n\}$, the return times are

$$\tau_{j,n} = t_{j+1} - t_j, \qquad j = 1, \ldots, N_n - 1.$$

These random intervals summarize recurrence to the stress region. Their empirical distribution captures spacing, dispersion, and asymmetry in a way that event counts cannot.

To describe clustering, fix a run parameter $r \geq 1$. A new cluster begins at time t if $X_t > u_n$ and the previous r observations do not exceed u_n. The corresponding finite-level cluster-start indicator is[1]

$$C_{t,n}^{(r)} = I_t(u_n) \prod_{j=1}^{r} \left(1 - I_{t-j}(u_n)\right).$$

The associated finite-level clustering coefficient is

$$\theta_n^{(r)} = \frac{\mathbb{E}[C_{0,n}^{(r)}]}{\mathbb{E}[I_0(u_n)]} = \mathbb{P}(X_{-1} \leq u_n, \ldots, X_{-r} \leq u_n \mid X_0 > u_n).$$

[1] The implementation uses a preceding-block definition. O'Brien's characterization is often written with a following block instead. For ergodic stationary sequences the two frequencies agree up to boundary terms, so the distinction is asymptotically negligible at the level used here.

When a $D^{(k)}$-type runs approximation is appropriate in the sense of O'Brien and subsequent declustering theory [15, 9], $\theta_n^{(r)}$ is a finite-level analogue of the extremal index: lower values indicate stronger local clustering of exceedances.

For a d-variate process $\{X_{t,j}\}$ with component-specific thresholds $u_{j,n}$, define

$$I_{t,j}(u_{j,n}) = \mathbf{1}\{X_{t,j} > u_{j,n}\}, \qquad S_t = \frac{1}{d}\sum_{j=1}^{d} I_{t,j}(u_{j,n}).$$

The scalar $S_t \in [0,1]$ is a multivariate stress-state index: it records the share of active stress coordinates at time t. It is descriptive, not causal, and it discards the richer extremal-dependence structure that would be described by objects such as stable tail dependence functions or tail-dependence coefficients.

4 Estimators and Implementation

4.1 Sample estimators

Given observations $X_1, \ldots, X_n$, the empirical exceedance rate at threshold u_n is

$$\widehat{p}_n(u_n) = \frac{1}{n}\sum_{t=1}^{n} I_t(u_n).$$

The runs-based estimator of the finite-level clustering coefficient is

$$\widehat{\theta}_n^{(r)} = \frac{\sum_{t=r+1}^{n} C_{t,n}^{(r)}}{\sum_{t=1}^{n} I_t(u_n)},$$

whenever the denominator is positive. This estimator is simple, transparent, and computationally appropriate for a reproducible pipeline. It is not always the most statistically efficient choice, but it is easy to audit and aligns with the article's emphasis on inspectable artifacts. Because the numerator starts at $t = r+1$ while the denominator counts exceedances over all t, exceedances in the first r positions can only bias the finite-sample estimate downward.

The empirical return-time distribution is

$$\widehat{G}_n(x) = \frac{1}{N_n - 1}\sum_{j=1}^{N_n-1} \mathbf{1}\{\tau_{j,n} \le x\},$$

provided $N_n \ge 2$. Summary functionals such as the mean, median, upper quantiles, and maximum of $\{\tau_{j,n}\}$ are used throughout the evidence section.

For the multivariate setting, the empirical stress-state index is

$$\widehat{S}_t = \frac{1}{d}\sum_{j=1}^{d} I_{t,j}(u_{j,n}),$$

and joint stress is defined by the event $\{\widehat{S}_t \geq 2/d\}$ in the present illustration. This threshold can be varied; the key point is to make the rule explicit, and in the present manuscript it should be read as a screening convention rather than as a structural threshold.

4.2 Threshold and run-length choices

Threshold selection is a substantive modeling decision, not a clerical preprocessing step. A threshold set too low destroys the rare-event interpretation; a threshold set too high produces severe finite-sample instability. The analysis therefore treats threshold sensitivity as part of the diagnostic output rather than as a hidden implementation detail. In applied work, one should report how clustering summaries move across a threshold grid and how the chosen run length r affects cluster identification.

The run parameter r has an equally direct interpretation. Larger values merge nearby exceedances into common clusters and therefore tend to increase measured persistence. This does not make large r "better." It means that r encodes what the analyst wishes to count as persistence. A serious stress paper should disclose that choice and test its sensitivity.

4.3 Repository implementation

The figures and source tables reported in this paper are generated by a deterministic implementation that proceeds from raw or simulated series to threshold definition, recurrence summaries, baseline econometric diagnostics, and article-facing outputs. The conceptual pipeline contains eight recorded stages: raw series, transformations, stress-region definition, exceedance events, clustering diagnostics, return-time diagnostics, econometric baselines, and article evidence. This separation matters because recurrence statistics are sensitive to thresholding and clustering rules; saved intermediate tables are therefore part of the method. Code and replication materials are available from the author.

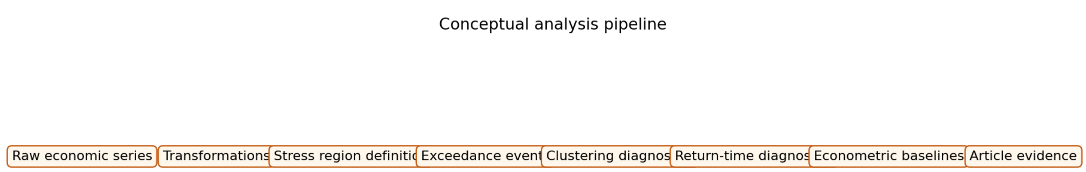

Figure 1: Conceptual pipeline for the computational workflow used in the paper. The methodological point is that recurrence objects are treated as explicit data products rather than informal visual impressions.

5 A Cluster-Adjusted Stress Functional

5.1 Definition

The paper's original theoretical contribution is deliberately narrow, but it is intended to be more than notational decoration. If recurrence is to be treated as part of stress measurement rather than as an auxiliary descriptive remark, then clustering information must be translated into a quantity that can be compared across processes and thresholding schemes. At level n, both the marginal tail probability and the clustering coefficient are finite-threshold objects, so the functional introduced here is

$$\Lambda_n(u_n, r) = \frac{p(u_n)}{\theta_n^{(r)}}.$$

Its sample analogue is

$$\widehat{\Lambda}_n(u_n, r) = \frac{\widehat{p}_n(u_n)}{\widehat{\theta}_n^{(r)}},$$

whenever $\widehat{\theta}_n^{(r)} > 0$.

The interpretation is straightforward only if the level of object is kept clear. The finite-threshold runs coefficient $\theta_n^{(r)}$ should be read here as a runs-based isolation coefficient, not as the literal reciprocal of mean cluster size. It records how isolated exceedances are relative to the declustering rule indexed by r, and at finite thresholds it depends on both the tail probability and the chosen run length. The functional $\Lambda_n(u_n, r)$ therefore does not claim to be an exact expected exceedance count or an exact probability. It is a stress functional that rescales exceedance incidence by lack of isolation, and for fixed $p(u_n)$ it increases when exceedances arrive in less isolated temporal configurations. This object is not a probability, and it can exceed one. The unemployment illustration below does exactly that, because the functional records cluster-weighted stress intensity rather than a frequency constrained to the unit interval. Because $\theta_n^{(r)} \leq 1$, the functional always satisfies $\Lambda_n(u_n, r) \geq p(u_n)$. At a finite threshold, however, the honest zero-clustering reference is not $p(u_n)$ itself but the independence benchmark

$$\Lambda_{\text{indep}}(u_n, r) = \frac{p(u_n)}{(1 - p(u_n))^r},$$

which exceeds $p(u_n)$ whenever $p(u_n) > 0$. This benchmark is the correct finite-threshold no-clustering reference for Λ, not a secondary comparison. In that sense, Λ may be read as a cluster-adjusted stress load: identical event frequencies need not imply identical stress severity if one process concentrates those events into more persistent bursts.

The functional is also non-injective in a way that matters empirically. A rare but strongly clustered regime and a more frequent but weakly clustered regime can generate the same value of Λ. For that reason, Λ should never be reported in isolation. The pair $(\widehat{p}_n, \widehat{\theta}_n^{(r)})$ must accompany it whenever substantive interpretation is attempted.

The simplicity of the definition is intentional. The aim is not to compete with the richer theory of tail processes or extremal dependence structures, but to produce a scalar summary that remains mathematically interpretable while being usable inside an applied stress workflow.

5.2 Compound-Poisson motivation

The definition above also has an asymptotic motivation, but that motivation should not be mistaken for an exact finite-threshold identification. In the standard compound-Poisson limit for exceedance point processes under dependence; see Leadbetter, Lindgren, and Rootzén [13] for the general framework and Hsing, Hüsler, and Leadbetter [12] for the exceedance-point-process formulation. Fix a threshold sequence u_n such that

$$n\,p(u_n) \to \tau \in (0, \infty),$$

and let

$$N_n = \sum_{t=1}^{n} \mathbf{1}\{X_t > u_n\}\,\delta_{t/n}$$

denote the normalized exceedance point process. Under the usual mixing and anti-clustering conditions, N_n converges to a compound Poisson process

$$N = \sum_{j=1}^{\infty} \xi_j \delta_{\Gamma_j},$$

where the cluster centers Γ_j form a Poisson process on $[0, 1]$ with rate $\theta\tau$ and the marks ξ_j are i.i.d. cluster sizes with mean $\mathbb{E}\xi_j = 1/\theta$.

That asymptotic picture explains why a clustering adjustment is natural: in the rare-event limit, smaller extremal index means that a given amount of tail mass is being delivered through fewer and larger episodes. What it does *not* justify on its own is identifying the finite-threshold runs coefficient $\theta_n^{(r)}$ with the reciprocal of literal mean cluster size. The asymptotic extremal index from the point-process limit and the finite-threshold runs-isolation coefficient used in the empirical workflow are related only heuristically in the present paper. For that reason, $\Lambda_n(u_n, r)$ should be read as a finite-threshold incidence functional motivated by clustered exceedance limits, not as a theorem-level identity for expected exceedance counts.

5.3 Basic properties

Lemma 1 (Elementary properties of the cluster-adjusted stress functional)**.** *Fix a threshold u and run length r. Consider two stationary processes A and B such that*

$$p_A(u) = p_B(u) = p(u) > 0, \qquad \theta_A^{(r)}(u) < \theta_B^{(r)}(u) \le 1.$$

Then

$$\Lambda_A(u, r) > \Lambda_B(u, r).$$

If exceedances are temporally independent at threshold u, then for every $r \ge 1$,

$$\theta^{(r)}(u) = (1 - p(u))^r \qquad \textit{and} \qquad \Lambda_{\text{indep}}(u, r) = \frac{p(u)}{(1 - p(u))^r}.$$

If $\{u_n\}$ is a high-threshold sequence with $p(u_n) \to 0$, then

$$\theta_n^{(r)} = (1 - p(u_n))^r \to 1 \qquad \text{and} \qquad \frac{\Lambda_{\text{indep}}(u_n, r)}{p(u_n)} \to 1.$$

Finally, if u_n is fixed and $1 \le r_1 < r_2$, then

$$\theta_n^{(r_2)} \le \theta_n^{(r_1)} \qquad \text{and} \qquad \Lambda_n(u_n, r_2) \ge \Lambda_n(u_n, r_1).$$

Proof sketch. The first claim follows immediately from the definition $\Lambda(u, r) = p(u)/\theta^{(r)}(u)$ and positivity of $p(u)$. Under independence,

$$\mathbb{P}(X_{-1} \le u, \ldots, X_{-r} \le u \mid X_0 > u) = \prod_{j=1}^{r} \mathbb{P}(X_{-j} \le u) = (1 - p(u))^r,$$

so the relevant finite-threshold benchmark is exactly the probability that the preceding r observations remain below threshold under temporal independence. Substituting this value into the definition of Λ gives the finite-level independence benchmark, and the asymptotic statement follows because $p(u_n) \to 0$ implies $(1 - p(u_n))^r \to 1$. If $r_2 > r_1$, then the event

$$\{X_{-1} \le u_n, \ldots, X_{-r_2} \le u_n\}$$

is a subset of

$$\{X_{-1} \le u_n, \ldots, X_{-r_1} \le u_n\},$$

so the monotonicity statement follows from conditional-probability ordering and the fixed numerator $p(u_n)$.

Lemma 2 (Relative plug-in consistency under imported ingredients)**.** *Assume:*

(A1) $\{X_t\}$ is strictly stationary and satisfies dependence conditions under which the runs estimator is consistent at intermediate thresholds;

(A2) u_n is an intermediate threshold sequence with $k_n = n\overline{F}(u_n) \to \infty$ and $k_n/n \to 0$;

(A3) for a fixed run length r, the finite-level clustering coefficient $\theta_n^{(r)}$ is positive and bounded away from zero eventually;

(A4) $\widehat{p}_n(u_n)/p(u_n) \xrightarrow{p} 1$ and $\widehat{\theta}_n^{(r)}/\theta_n^{(r)} \xrightarrow{p} 1$.

Then

$$\frac{\widehat{\Lambda}_n(u_n, r)}{\Lambda_n(u_n, r)} \xrightarrow{p} 1, \qquad \text{equivalently} \qquad \widehat{\Lambda}_n(u_n, r)\frac{\theta_n^{(r)}}{p(u_n)} \xrightarrow{p} 1.$$

Proof sketch. Write

$$\frac{\widehat{\Lambda}_n(u_n, r)}{\Lambda_n(u_n, r)} = \frac{\widehat{p}_n(u_n)}{p(u_n)} \cdot \frac{\theta_n^{(r)}}{\widehat{\theta}_n^{(r)}}.$$

The first factor converges to one by assumption. The second does as well because $\widehat{\theta}_n^{(r)}/\theta_n^{(r)} \xrightarrow{p} 1$ and the reciprocal map is continuous on $(0, \infty)$. The continuous mapping theorem therefore yields the relative consistency claim. The purpose of the lemma is bookkeeping rather than originality: it does not prove consistency of the runs estimator. That ingredient is imported from the extremal-index literature under mixing and intermediate-threshold conditions; see, for example, [9, 16].

Proposition 1 (Finite-sample perturbation bound)**.** *Let $p > 0$ and $\theta > 0$, and let $\widetilde{p}$ and $\widetilde{\theta}$ be estimators such that*

$$|\widetilde{p} - p| \leq \varepsilon_p, \qquad |\widetilde{\theta} - \theta| \leq \varepsilon_\theta < \theta.$$

Then

$$\left|\frac{\widetilde{p}}{\widetilde{\theta}} - \frac{p}{\theta}\right| \leq \frac{\varepsilon_p}{\theta - \varepsilon_\theta} + \frac{p\,\varepsilon_\theta}{\theta(\theta - \varepsilon_\theta)}.$$

Proof sketch. Write

$$\frac{\widetilde{p}}{\widetilde{\theta}} - \frac{p}{\theta} = \frac{\theta(\widetilde{p} - p) + p(\theta - \widetilde{\theta})}{\theta\widetilde{\theta}}.$$

Taking absolute values and using $\widetilde{\theta} \geq \theta - \varepsilon_\theta > 0$ gives

$$\left|\frac{\widetilde{p}}{\widetilde{\theta}} - \frac{p}{\theta}\right| \leq \frac{\theta\varepsilon_p + p\varepsilon_\theta}{\theta(\theta - \varepsilon_\theta)},$$

which is the stated bound after separating terms. The interpretation is important: estimation error in the clustering term is amplified when θ is small, so cluster-adjusted stress is most numerically delicate precisely in the regimes where clustering is strongest.

Remark 1. *The functional $\Lambda(u_n, r)$ is best understood as a ranking and summary device, not as a welfare object or a structural sufficient statistic. Its value lies in comparative stress characterization.*

Remark 2. *The perturbation bound explains why threshold sensitivity cannot be treated as a cosmetic appendix item. If threshold changes move $\widehat{\theta}_n^{(r)}$ toward zero, the same sampling error in the clustering estimator induces a larger error in $\widehat{\Lambda}_n$. This is a concrete mathematical reason to report threshold and run-length robustness.*

6 Synthetic Evidence

6.1 Core numerical outputs

Table 1 collects the generated quantities used in the analysis.

Table 1: Core artifact summaries

Object	Numerical summary
Finite-level clustering coefficient by series (artifact 04; runs / intervals estimates for clustered synthetic, isolated synthetic, logistic benchmark observable)	Clustered synthetic stress: 0.673/0.721; isolated synthetic stress: 0.780/0.988; logistic benchmark observable: 1.000/1.000
Return-time diagnostics (artifact 03; logistic calibration in source table, clustered synthetic benchmark, Baa spread)	Logistic calibration: $\widehat{\theta}_{\text{runs}} = 1.000$, $\widehat{\theta}_{\text{int}} = 1.000$, CV 0.795; clustered stress: 0.673/0.721, CV 1.286; Baa spread: 0.186/0.213, CV 2.417
Multivariate stress timeline (artifact 05; coupled synthetic panel)	120 dated observations; 12 joint exceedance dates; maximum of 2 active series; mean stress score 0.150
Volatility comparison (artifact 06; macro-stress baseline series)	Mean rolling volatility during exceedances 0.3429; outside exceedances 0.2671; exceedance count 90

These quantities do not appear here as decorative summaries appended to a computational pipeline. They define the evidentiary boundary of the paper. Every substantive claim that follows is tied either to these numerical outputs or to the saved source tables from which the corresponding figures are drawn. That restriction is methodologically important, because the manuscript is intended to argue from inspectable evidence rather than from narrative inflation.

6.2 Synthetic clustering benchmarks

The clearest evidence for the paper's thesis comes from the finite-level clustering-coefficient comparison between the two synthetic stress benchmarks. The clustered benchmark yields an estimated coefficient of 0.673, whereas the more isolated benchmark yields 0.780. The numerical gap is not large enough to justify dramatic rhetoric, and the manuscript should resist the temptation to manufacture significance through tone. Its importance lies elsewhere. The ranking shows that recurrence diagnostics recover a difference in temporal organization that would be awkward to express, and even easier to ignore, if the discussion remained confined to generic language about instability or turbulence.

This is also the context in which the cluster-adjusted functional acquires substantive meaning rather than remaining a formal convenience. For comparable exceedance incidence, the series with the smaller clustering coefficient receives the larger stress load, because the same overall amount of tail activity is being delivered in a more temporally concentrated way. The theoretical section is therefore not separate from the synthetic evidence; it explains why the empirical ranking should matter in the first place.

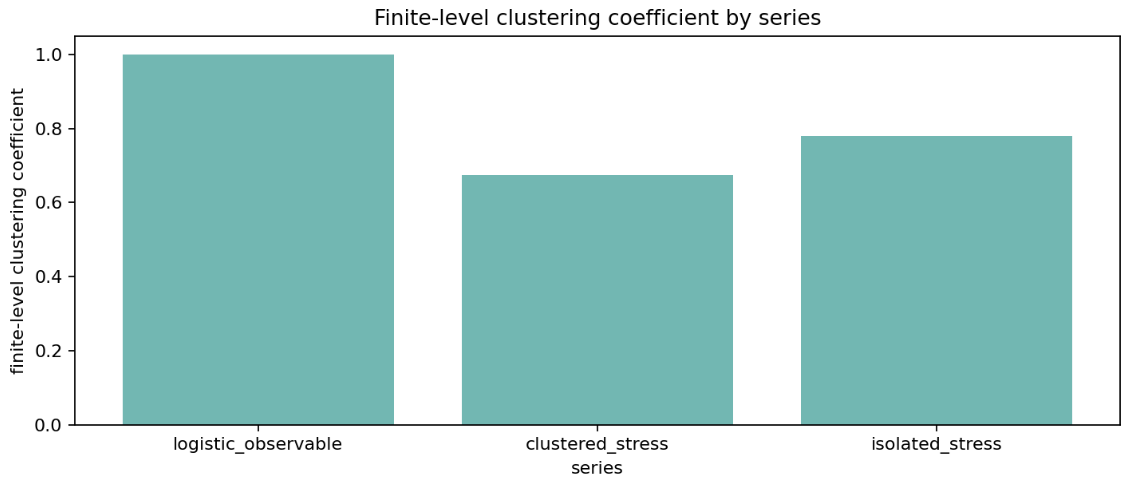

Figure 2: Estimated finite-level clustering coefficients across benchmark series. Lower values indicate stronger local clustering of exceedances. The result is finite-sample and threshold-dependent, but it recovers the intended ranking between clustered and more isolated stress processes.

6.3 Threshold robustness and uncertainty

Any recurrence-based stress diagnostic inherits two forms of uncertainty at once. The first is ordinary sampling uncertainty: finite samples generate noisy exceedance counts, noisy cluster counts, and therefore noisy clustering summaries. The second is design uncertainty: the reported statistic depends on threshold choice and, for runs-based procedures, on the analyst's declustering rule. For that reason, robustness over a threshold grid is not a cosmetic appendix exercise but part of the object being measured. The analysis below makes that point operational by pairing the threshold scan with a moving-block bootstrap band, so that the threshold path is accompanied by a finite-sample uncertainty envelope rather than by point estimates alone.

The threshold-sensitivity figure makes this point directly. The threshold scan uses run length $r = 4$ on the logistic benchmark observable

$$Y_t = -\log\bigl(|X_t - 1/2| + 10^{-12}\bigr), \qquad X_{t+1} = 4X_t(1 - X_t).$$

The distinguished point 1/2 is not periodic for the fully chaotic logistic map, so the periodic-point mechanism that can force extremal indices below one in dynamical-systems EVT is absent here [10]. The benchmark is therefore supposed to behave like a near-null calibration object. In that scan, the runs-based clustering estimate is 0.888 at the 0.90 quantile with 500 exceedances and rises to 0.909 at the 0.93 quantile with 350 exceedances. By the 0.95 quantile, the estimate reaches 1.000 with 250 exceedances and remains at that level through the 0.97 and 0.99 quantiles, even as the number of exceedances falls to 150 and then 50. Once the finite-level independence baseline is added, the correct reading becomes more precise. The corresponding i.i.d. benchmark values are $(1 - \widehat{p})^4 = 0.656$, 0.748, 0.815, 0.885, and 0.961, so the entire path lies above independence rather than below it. In other words, this scan is best read as a null-calibration exercise on a benchmark observable whose upper tail is already close to the finite-level no-clustering reference, not as direct evidence of persistent tail bunching.

That benchmark comparison is cleaner if the clustering estimate is renormalized directly. Define

$$\widetilde{\theta}_n^{(r)} = \frac{\widehat{\theta}_n^{(r)}}{(1-\widehat{p}_n)^r}.$$

Under finite-threshold independence, the runs benchmark is exactly $(1-\widehat{p}_n)^r$, so $\widetilde{\theta}_n^{(r)} = 1$ is the correct no-clustering reference at every threshold and run length. This is why $\widetilde{\theta}$ is a more honest cross-design comparison object than $\widehat{\theta}$ alone. In the logistic threshold scan, the renormalized values are 1.353, 1.216, 1.228, 1.130, and 1.041 across the 0.90, 0.93, 0.95, 0.97, and 0.99 quantiles. The path therefore confirms the same interpretation from a different angle: the benchmark observable is not showing excess clustering relative to independence. If anything, its finite-threshold exceedances are slightly more isolated than the exact i.i.d. runs benchmark, and that gap narrows as the threshold tightens.

This threshold path also clarifies what uncertainty quantification should mean in a completed empirical application. A single point estimate of $\widehat{\theta}_n^{(r)}$ or $\widehat{\Lambda}_n(u_n, r)$ is not enough, because inferential uncertainty and threshold sensitivity interact. The bootstrap-enhanced figures and source tables are a first step in the right direction: they do not solve the inferential problem, but they do allow the analyst to see whether apparent stability at a given threshold reflects robust clustering evidence or simply the collapse of effective sample size. The same logic now extends to the cluster-adjusted stress functional itself. In the current threshold scan, the estimated functional falls from 0.113 at the 0.90 quantile to 0.077 at 0.93, then to 0.050, 0.030, and 0.010 at the 0.95, 0.97, and 0.99 quantiles, respectively, while the corresponding finite-level independence benchmark falls from 0.152 to 0.094, 0.061, 0.034, and 0.010. The associated 90% block-bootstrap intervals are $[0.111, 0.118]$, $[0.076, 0.080]$, $[0.0498, 0.0512]$, $[0.0298, 0.0306]$, and $[0.0096, 0.0102]$. The important point is not merely that $\widehat{\Lambda}$ declines. It is that the decline largely tracks the finite-threshold mechanical benchmark, which is exactly what one should expect from a null-calibration observable with little excess clustering beyond the $(1-\widehat{p})^r$ effect.

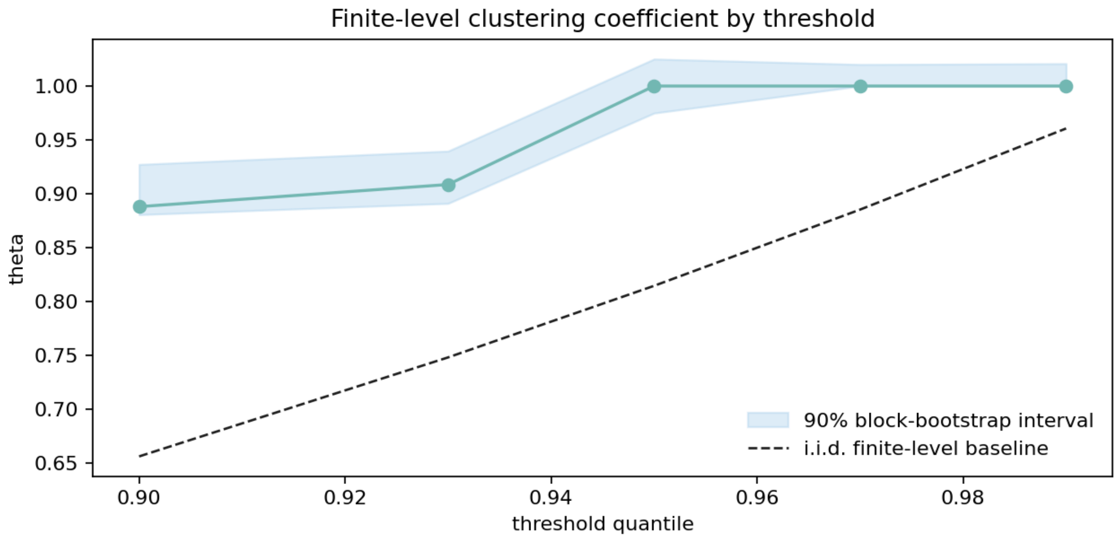

Figure 3: Threshold sensitivity of the clustering estimate. The dashed curve is the finite-level i.i.d. benchmark $(1-\widehat{p})^r$, so the figure shows not only how the estimate moves with the threshold, but also whether it departs materially from the no-clustering reference at each cutoff.

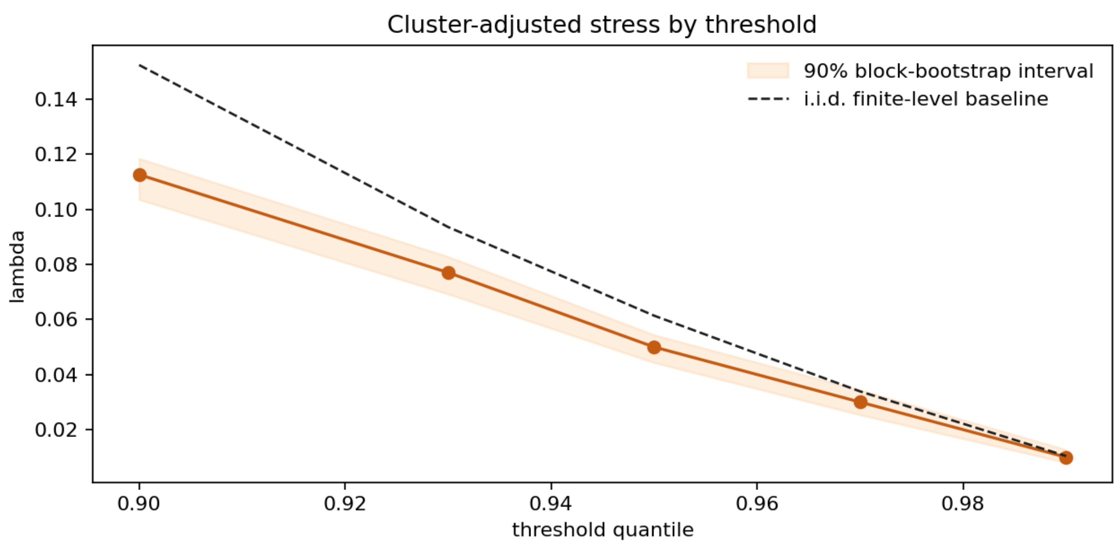

Figure 4: Threshold sensitivity of the cluster-adjusted stress functional. The dashed curve is the finite-level independence benchmark $\widehat{p}/(1-\widehat{p})^r$, which is the honest reference for deciding whether the observed path reflects excess clustering or merely the mechanical $(1-\widehat{p})^r$ adjustment.

Run-length sensitivity provides a second robustness check because the declustering rule is itself part of the stress definition. At the fixed 0.90 threshold quantile used in the run-length scan on the same logistic benchmark observable, the estimated cluster-adjusted functional equals 0.100 for run lengths 1 and 2, then rises to 0.113, 0.149, and 0.206 for run lengths 4, 6, and 8, respectively. The corresponding 90% block-bootstrap intervals are $[0.0998, 0.1006]$, $[0.1002, 0.1012]$, $[0.1115, 0.1174]$, $[0.1445, 0.1591]$, and $[0.1968, 0.2213]$. Taken on their own, those numbers might invite a persistence interpretation, but that reading is false. The finite-level independence benchmark at $\widehat{p} = 0.10$ is $\widehat{\Lambda}_{\text{indep}} = 0.111$, 0.123, 0.152, 0.188, and 0.233 for $r = 1$, 2, 4, 6, and 8, respectively, so the observed path sits at or below independence throughout. The increase in $\widehat{\Lambda}$ with r is therefore mechanical rather than diagnostic: it occurs under i.i.d. data as well because $(1 - \widehat{p})^r$ falls with r. The substantive quantity is the gap to the independence reference, not raw monotonicity in the run length.

The renormalized clustering path makes the same point with less room for confusion. At the same threshold, $\widetilde{\theta}$ equals 1.111, 1.220, 1.348, 1.264, and 1.130 for $r = 1$, 2, 4, 6, and 8. Those values are not constant, so renormalization does not magically remove all finite-sample variation, but it does remove the false intuition that a rising raw $\widehat{\Lambda}$ or a falling raw $(1 - \widehat{p})^r$ is automatically evidence of more persistence. The benchmark observable stays above one throughout, which is exactly what the earlier threshold scan already suggested: this is a calibration object whose exceedances are not more tightly bunched than the finite-level independence reference.

Figure 5: Run-length sensitivity of the cluster-adjusted stress functional at the 0.90 threshold quantile. The dashed curve is the finite-level i.i.d. benchmark, so the figure makes clear that raw monotonicity in r carries no persistence information by itself; only departures from the benchmark do.

Table 2: Compact robustness summary for the cluster-adjusted stress functional

Path	Setting	$\widehat{\Lambda}$	$\widehat{\Lambda}_{\text{indep}}$	90% bootstrap interval
Threshold	$q = 0.90$	0.113	0.152	[0.111, 0.118]
Threshold	$q = 0.93$	0.077	0.094	[0.076, 0.080]
Threshold	$q = 0.95$	0.050	0.061	[0.0498, 0.0512]
Threshold	$q = 0.97$	0.030	0.034	[0.0298, 0.0306]
Threshold	$q = 0.99$	0.010	0.010	[0.0096, 0.0102]
Run length	$r = 1$	0.100	0.111	[0.0998, 0.1006]
Run length	$r = 2$	0.100	0.123	[0.1002, 0.1012]
Run length	$r = 4$	0.113	0.152	[0.1115, 0.1174]
Run length	$r = 6$	0.149	0.188	[0.1445, 0.1591]
Run length	$r = 8$	0.206	0.233	[0.1968, 0.2213]

6.4 Return-time diagnostics through the Ferro–Segers mixture

The return-time section is sharper once it is organized around the same clustering object used elsewhere in the paper. Under local dependence, the normalized inter-exceedance times do not simply target a unit exponential law. Their limiting benchmark is the Ferro–Segers mixture

$$(1-\theta)\delta_0 + \theta\,\mathrm{Exp}(\theta),$$

whose atom at zero records the share of within-cluster recurrences and whose exponential tail describes the spacing between clusters themselves. In the present paper that law is used as a fitted diagnostic approximation, not as a claim about an exact finite-sample distribution. That formulation is more disciplined than talking vaguely about "wide waiting-time distributions," because it ties the return-time evidence directly to the extremal-index discussion rather than treating it as a separate descriptive aside.

The logistic benchmark observable remains useful precisely because it collapses back to the null-like case. At threshold quantile 0.95 with run length $r = 4$, both the runs estimator and the intervals estimator equal 1.000. The artifact contains 249 inter-exceedance intervals, the normalized coefficient of variation is 0.795, the KS distance from the unit exponential benchmark is 0.221, and the moving-block bootstrap reference value is 0.056. Read properly, that panel is not supposed to be dramatic. It is a calibration object showing what the mixture looks like when the atom effectively disappears and the recurrence pattern is close to the no-clustering reference.

The clustered synthetic benchmark is the more revealing case. Using the absolute GARCH$(1,1)$ benchmark at the same threshold quantile and run length gives $\widehat{\theta}_{\text{runs}} = 0.673$ and $\widehat{\theta}_{\text{int}} = 0.721$, together with 149 return times, a median waiting time of 10.0, a 90th percentile of 53.6, a maximum of 145, and a normalized coefficient of variation of 1.286. The gap between the two estimators is small enough to be reassuring rather than troublesome: both identify substantial local clustering. Once the QQ panel is read against the fitted mixture instead of against a pure exponential line, its geometry becomes more interpretable. Excess short waits correspond to repeated exceedances arriving inside the same burst, while the stretched right tail reflects the quiet spells separating one burst from the next.

The empirical Baa–Treasury spread sits between the logistic calibration and the deliberately clustered synthetic process, which is exactly why it is a useful second panel. At threshold quantile 0.90 with run length $r = 3$, the series yields $\widehat{\theta}_{\text{runs}} = 0.186$ and $\widehat{\theta}_{\text{int}} = 0.213$, so the fitted mixture places atom mass $1 - \widehat{\theta}_{\text{int}} = 0.787$ at zero. Its normalized coefficient of variation is 2.417, much larger than in the logistic calibration and larger even than in the synthetic clustered benchmark. I do not want to over-interpret a single monthly spread series, but the point of the panel is narrower than that: the same mixture language remains readable outside simulation, and it describes an empirical stress process with short bunches of exceedances separated by long quieter stretches.

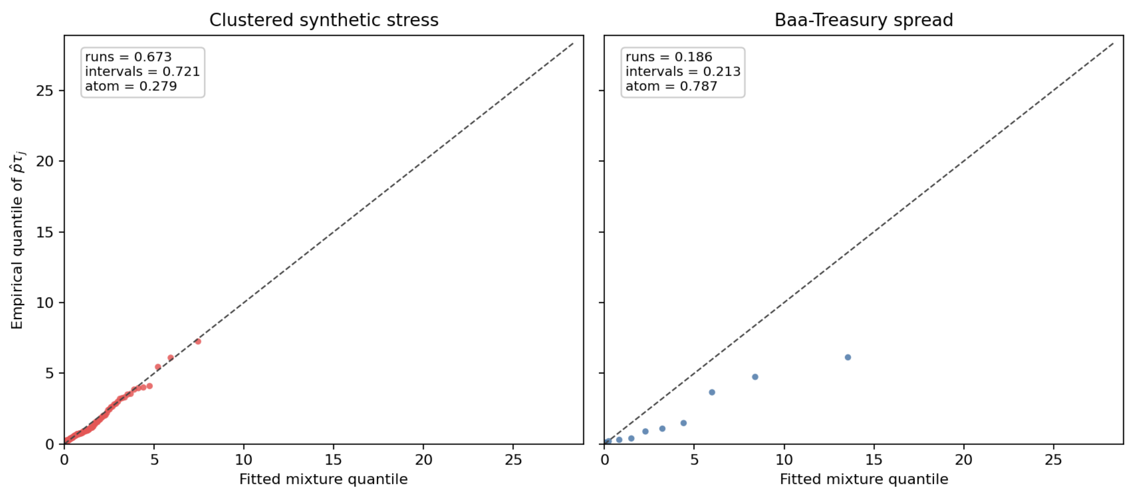

Figure 6: QQ comparison of normalized return times $\widehat{p}\tau_j$ against the fitted Ferro–Segers mixture. The clustered synthetic benchmark shows the expected excess of short recurrence gaps and a stretched right tail, while the Baa–Treasury spread exhibits the same geometry in milder but still visible form.

6.5 Multivariate stress activation

The multivariate illustration records 120 observations and 12 joint exceedance dates, with at most two simultaneously active series and a mean stress score of 0.150. The claim supported by these numbers is intentionally modest, but it is still consequential. Stress is not uniformly distributed across coordinates through time: some episodes are effectively univariate, while others involve simultaneous activation across variables. That descriptive distinction should be measured before stronger language about contagion, propagation, or systemic amplification is introduced.

The stress-state index is therefore best understood as a screening device for joint activation rather than as a substitute for structural multivariate analysis. Its purpose is to mark the temporal geometry of shared stress, not to explain why that geometry arises.

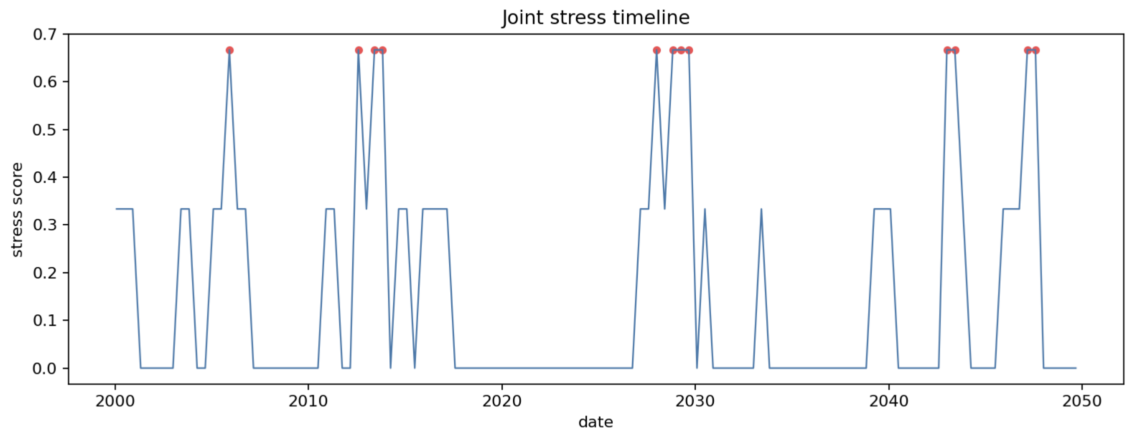

Figure 7: Multivariate stress-state timeline. The figure does not identify transmission channels; it records when stress is isolated and when it is joint across observables.

6.6 Why the volatility baseline remains necessary

The baseline comparison shows higher mean rolling volatility during exceedance periods (0.3429) than outside them (0.2671). This is both substantively useful and entirely unsurprising. A recurrence framework should not be judged by whether it overturns established volatility evidence, because that is not the standard it is meant to meet. Its purpose is to recover information that volatility summaries do not encode directly.

The proper conclusion is therefore one of complementarity rather than rivalry. Volatility remains the natural object when the question concerns changing scale conditions, while recurrence diagnostics become essential when the question concerns how extreme states reappear and bunch together through time. A stress workflow that excludes either object is, in different ways, incomplete.

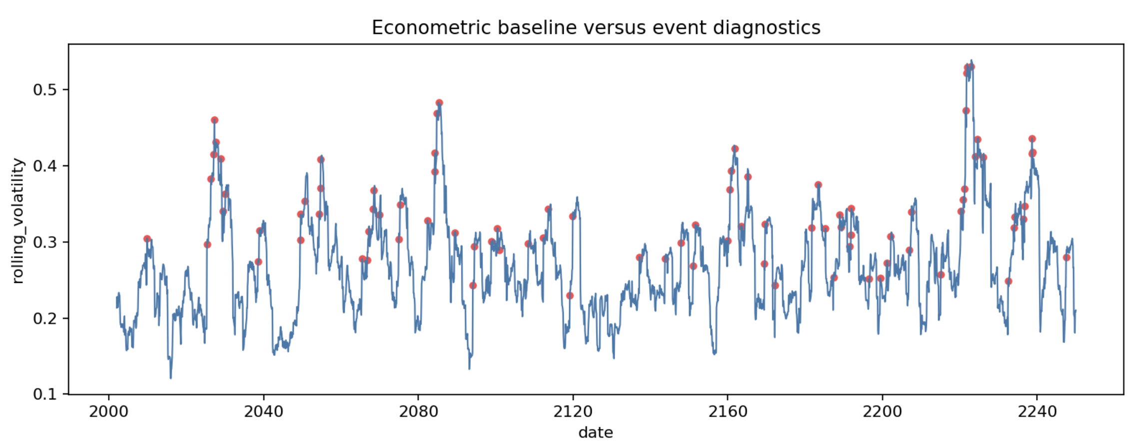

Figure 8: Baseline econometric and recurrence-oriented diagnostics are complementary. Elevated volatility during exceedance periods is informative, but it does not encode clustering intensity or return-time dispersion.

6.7 A small empirical macro-financial panel

The paper now includes a genuine, if still modest, empirical illustration built from three publicly available monthly FRED series rather than from a toy bundled example. The panel consists of the civilian unemployment rate, the Moody's Baa minus 10-year Treasury spread, and the Kansas City Financial Stress Index. After aligning the cleaned local CSV extracts on their common support, the usable sample runs from February 1990 through May 2026 and contains 435 monthly observations. For comparability with the synthetic diagnostics, the baseline empirical specification uses upper-tail exceedances at the 0.90 quantile together with a runs declustering parameter of $r = 3$. Because the unemployment level is visibly dominated by low-frequency regime shifts, I also report a 12-month difference for that series; the transformation removes slow-moving level changes while preserving a monthly cyclical interpretation.

At that baseline specification, the unemployment level row is best treated as descriptive only. Its threshold is 8.2; it produces 42 exceedances, but those exceedances collapse into only 2 runs clusters, yielding $\widehat{\theta}_{\text{runs}} = 0.048$, $\widehat{\theta}_{\text{int}} = 0.049$, and $\widehat{\Lambda} = 2.028$. Those numbers mostly restate the obvious historical fact that the sample contains two long recessionary unemployment episodes, and the stationarity and mixing assumptions from Section 3 are not credible enough on the level scale to support strong inferential language. The transformed unemployment series is more informative for that purpose. At the same threshold quantile and run length, the 12-month difference has threshold 1.30, 42 exceedances, 4 runs clusters, $\widehat{\theta}_{\text{runs}} = 0.095$, $\widehat{\theta}_{\text{int}} = 0.142$, and $\widehat{\Lambda} = 1.043$. The two financial indicators remain less concentrated by point estimate: the Baa spread crosses its threshold of 3.16 in 43 months grouped into 8 clusters, giving $\widehat{\theta}_{\text{runs}} = 0.186$, $\widehat{\theta}_{\text{int}} = 0.213$, and $\widehat{\Lambda} = 0.531$, while KCFSI crosses its threshold of 1.068 in 44 months also grouped into 8 clusters, giving $\widehat{\theta}_{\text{runs}} = 0.182$, $\widehat{\theta}_{\text{int}} = 0.161$, and $\widehat{\Lambda} = 0.556$. The empirical message is therefore narrower than the earlier draft implied. Level unemployment stress is concentrated in a few long blocks, but the inferentially usable comparison is between the transformed unemployment series and the financial indicators, where the distinction remains present but no longer supports a dramatic ordinal claim. The intervals estimator now appears beside the runs estimator because it offers a second reading of the same recurrence geometry; here it slightly enlarges the transformed unemployment and Baa estimates, lowers KCFSI modestly, and leaves the substantive ranking essentially intact.

The bootstrap treatment has also been corrected. The earlier percentile intervals mixed threshold re-selection into what was supposed to be fixed-design sampling uncertainty, and for low-cluster cases they produced exactly the kind of boundary pathology that should have triggered rejection rather than reporting. The revised empirical intervals condition on the original numeric threshold, use basic bootstrap reflection, report a sensitivity envelope over block sizes $\{12, 24, 36, 48\}$ rather than a single ad hoc block choice, and declare rows with fewer than 3 clusters not interpretable instead of forcing a numeric band. Those envelopes are built from nominal 90% block-bootstrap intervals, but the envelope itself should not be read as carrying a literal 90% coverage guarantee once the outer band across block sizes is taken. That is why the unemployment level row in Table 3 now carries an explicit degeneracy note rather than a misleading confidence band. For the transformed unemployment series, the sensitivity envelopes built from 90% block-bootstrap intervals are $[0, 0.190]$ for $\widehat{\theta}_{\text{runs}}$ and $[0, 1.934]$ for $\widehat{\Lambda}$. For the Baa spread, they are $[0.002, 0.213]$ and $[0.093, 0.903]$; for KCFSI, $[0, 0.267]$ and $[0, 1.043]$. These are wide bands, and they should be read that way. The transformed unemployment point estimate remains larger than the two financial estimates, but interval overlap is substantial, so the uncertainty treatment supports caution rather than a hard ranking.

A small synthetic coverage harness was added to check that the repaired interval construction is at least numerically coherent on the clustered benchmark used elsewhere in the paper. Relative to a long-run benchmark reference with $\theta \approx 0.646$ and $\Lambda \approx 0.155$, the nominal 90% basic bootstrap intervals achieved empirical coverage of 0.875 for θ and 1.000 for Λ across 40 replications. That is not a proof of finite-sample validity, but it is enough to show that the revised procedure is no longer failing the elementary self-checks that the earlier intervals did.

Table 3: Empirical clustering summary for the monthly FRED panel, with sensitivity envelopes built from block-bootstrap intervals

Series	$\widehat{\theta}_{\text{runs}}$	$\widehat{\theta}_{\text{int}}$	sensitivity envelope	$\widehat{\Lambda}$	sensitivity envelope
Unemployment rate (level)	0.048	0.049	not interpretable ($k = 2$)	2.028	not interpretable ($k = 2$)
Unemployment rate (12m diff.)	0.095	0.142	[0, 0.190]	1.043	[0, 1.934]
Baa-Treasury spread	0.186	0.213	[0.002, 0.213]	0.531	[0.093, 0.903]
KCFSI	0.182	0.161	[0, 0.267]	0.556	[0, 1.043]

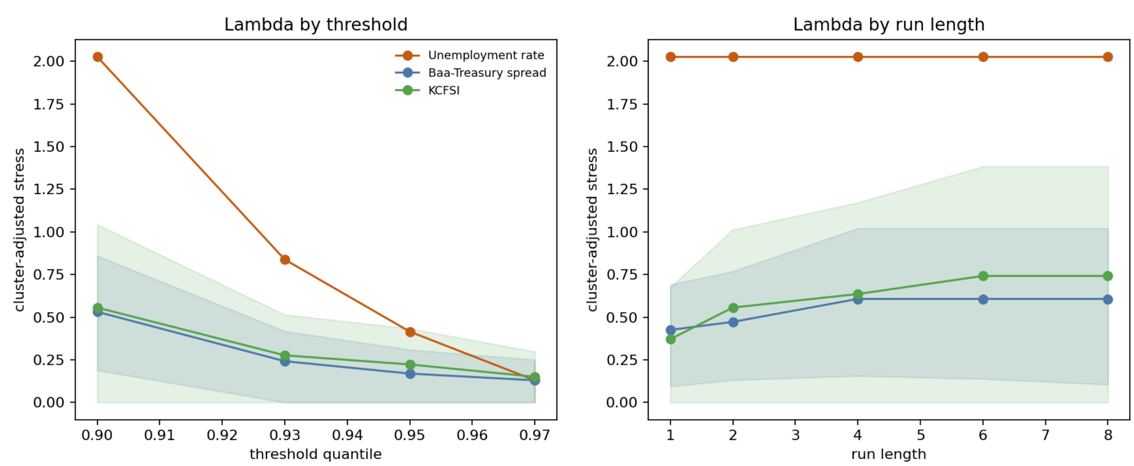

Figure 9: Empirical robustness of the cluster-adjusted stress load. The level-unemployment path should be read descriptively because it is generated by only two baseline clusters; the financial indicators remain the more interpretable comparison objects under threshold and run-length variation.

The robustness figure is still useful, but its interpretation must now respect the inferential split. Along the threshold dimension, level unemployment is the most sensitive series: its estimated stress load falls from 2.028 at $q = 0.90$ to 0.838, 0.415, and 0.130 at $q = 0.93$, 0.95, and 0.97, respectively. That compression is descriptive evidence that the broader upper-tail unemployment regime is dominated by a few long episodes, whereas the most extreme observations are less tightly packed. The Baa spread and KCFSI decline more smoothly, moving from 0.531 to 0.130 and from 0.556 to 0.150 over the same threshold grid. Along the run-length dimension, level unemployment is essentially invariant because its two major stress episodes are already separated by long calm intervals, while the two financial indicators become progressively more clustered as the episode definition widens. For the Baa spread, $\widehat{\Lambda}$ rises from 0.425 at $r = 1$ to 0.607 by $r = 4$; for KCFSI it rises from 0.371 at $r = 1$ to 0.742 by $r = 6$. The transformed unemployment specification is not plotted in that figure because its purpose is inferential repair rather than visual comparison, but its baseline point estimate sits between the level-unemployment and financial-series readings. The corrected conclusion is therefore conditional: labor-market stress is the most concentrated object descriptively in levels, yet the inferentially usable differences against the financial indicators are materially less sharp once the transformation and interval repair are imposed.

The renormalized clustering diagnostic sharpens that reading further. Because $\widetilde{\theta} = 1$ under independence at every finite threshold and run length, values below one indicate more bunching than the exact i.i.d. benchmark and values above one indicate more isolation. At the baseline empirical specification $(q, r) = (0.90, 3)$, the renormalized runs estimates are 0.065 for unemployment in levels, 0.130 for transformed unemployment, 0.254 for the Baa spread, and 0.250 for KCFSI. That ranking is more stable than the raw $\widehat{\Lambda}$ ranking because it removes the mechanical effect of changing $(1 - \widehat{p})^r$ across designs.

It also answers the run-length question more cleanly than the raw robustness plot. The earlier visual pattern suggested a reversal: level unemployment looked flat in r, whereas the financial indicators rose as the declustering window widened. That statement does not survive renormalization literally. Once the exact independence benchmark is divided out, the level-unemployment path is no longer flat at all; it rises from 0.053 to 0.058, 0.071, 0.088, and 0.107 as r moves from 1 to 2, 4, 6, and 8. The Baa spread takes values 0.258, 0.258, 0.247, 0.304, and 0.375, while KCFSI takes 0.303, 0.225, 0.244, 0.258, and 0.320. So the sharp raw reversal disappears. What remains is the more substantive statement: all three series stay below one throughout, and unemployment remains the most clustered object relative to the finite-threshold independence benchmark even after the mechanical run-length effect is removed.

Table 4: Compact empirical robustness summary for the cluster-adjusted stress load

Series	Threshold path $\widehat{\Lambda}(q, 3)$	Run-length path $\widehat{\Lambda}(0.90, r)$
Unemployment rate	$2.028 \to 0.838 \to 0.415 \to 0.130$	$2.028 \to 2.028 \to 2.028 \to 2.028 \to 2.028$
Baa-Treasury spread	$0.531 \to 0.242 \to 0.169 \to 0.130$	$0.425 \to 0.472 \to 0.607 \to 0.607 \to 0.607$
KCFSI	$0.556 \to 0.276 \to 0.223 \to 0.150$	$0.371 \to 0.556 \to 0.636 \to 0.742 \to 0.742$

Joint activation sharpens the historical interpretation. Defining a joint stress month as one in which at least two of the three indicators exceed their respective thresholds, the panel contains 29 such months and an average stress score of 0.099, with all three indicators active simultaneously at the peak. The dominant cluster runs from August 2008 through August 2009, which is precisely the kind of prolonged, multi-margin stress episode the recurrence framework is meant to distinguish from short-lived turbulence. Smaller joint clusters appear around the 2001 recession, the 2011 euro-area stress period, and the initial pandemic shock in March through May 2020. The empirical point is not that these dates are surprising; it is that the framework measures them in a way that keeps incidence, clustering, and joint activation analytically separate.

This remains a restrained empirical section rather than a full application. The thresholds are deliberately simple, no structural identification is attempted, and the evidence should not be read as a claim about optimal early-warning design. Even so, the illustration is now doing the right kind of work for a methods paper. It shows that recurrence diagnostics survive contact with an interpretable macro-financial panel, but also that they force discipline about when a dramatic point estimate is only descriptive and when an inferential comparison is actually defensible. The main empirical distinction remains the contrast between long labor-market stress blocks and more episodic, design-sensitive financial turbulence, but the uncertainty repair makes that contrast narrower and more credible.

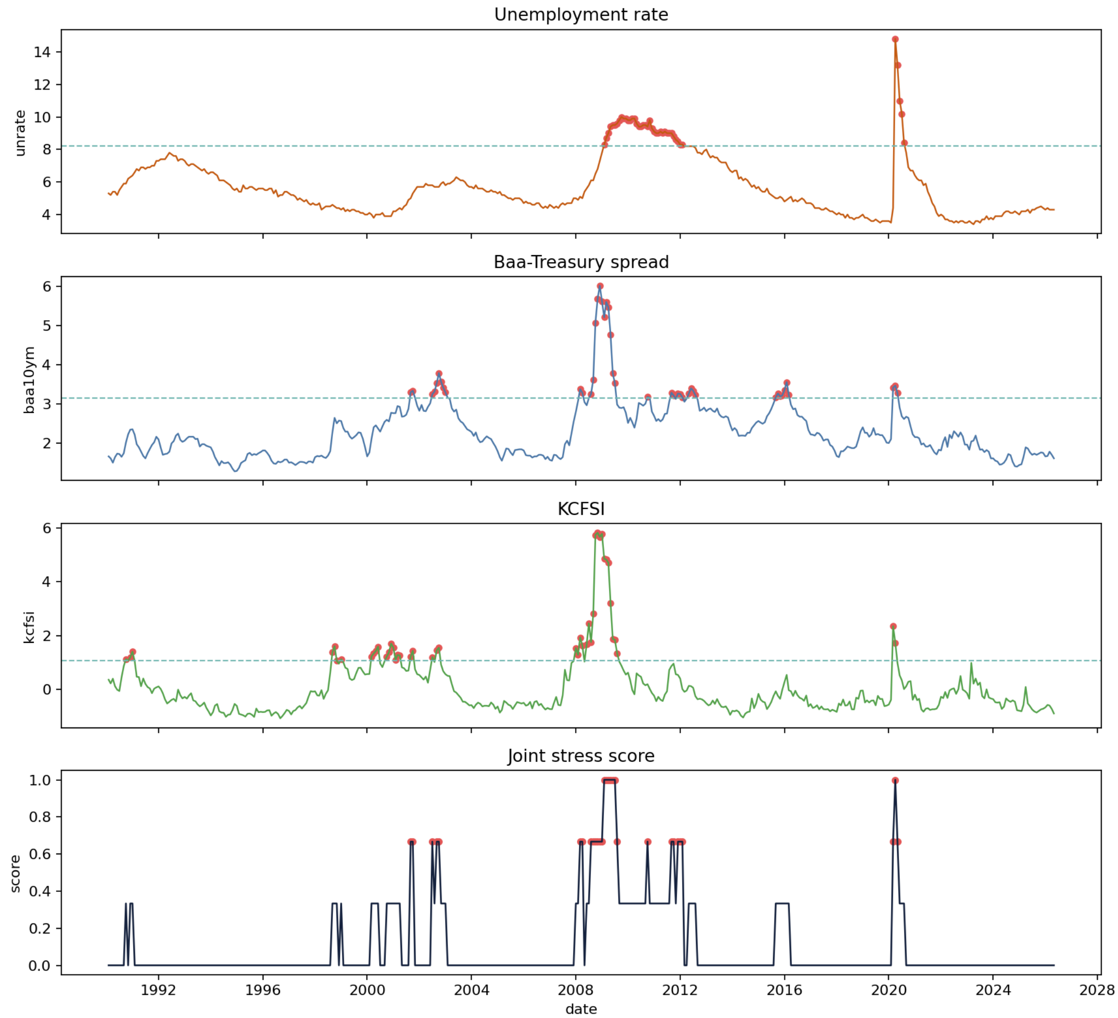

Figure 10: Empirical macro-financial illustration from the monthly FRED panel. The unemployment level exhibits highly concentrated upper-tail stress in a descriptive sense, while the credit spread and the financial-stress index recur across more separate episodes; the joint stress score peaks during the 2008–2009 crisis cluster.

7 Limitations and Scope

The limits of the paper are easiest to understand once its contribution is stated without inflation. The theoretical results are intentionally narrow. They establish interpretable properties of the cluster-adjusted stress functional and show how its behavior depends on the clustering term, but they do not amount to a new asymptotic theory for extremal-index estimation, a solution to threshold selection, or a general finite-sample theory for recurrence diagnostics. The paper therefore contributes an organizing functional and a set of elementary results around it, not a replacement for the deeper probabilistic literature on dependent extremes.

The empirical limits are equally important. The evidence presented here is primarily synthetic, which is appropriate for showing what the diagnostics do and do not measure, but insufficient for strong claims about macro-financial systems in the wild. A more mature empirical paper would need defended threshold rules, uncertainty quantification, robustness analysis across run lengths and sample windows, and an interpretation anchored in a specific domain rather than in generic language about stress. Without those additions, the present manuscript should be read as methodological preparation for empirical work rather than as empirical resolution.

The multivariate component is narrower still. The stress-state index records when threshold-based stress is isolated and when it is joint, but it does not identify contagion, directional spillovers, or structural transmission channels. That limitation is not incidental. It marks the boundary between descriptive recurrence diagnostics and the larger enterprise of systemic-risk modeling.

These constraints should not be treated as apologetic caveats appended after the fact. They are part of the paper's intellectual discipline. The contribution is a methods and theory bridge for recurrence diagnostics in stress analysis, and it is more useful to state that contribution clearly than to overstate what the present evidence can bear.

8 Conclusion

The central argument of the paper is that stress analysis must distinguish tail magnitude from tail organization. Econometric and risk-measure literatures have developed powerful tools for the first object, but the second requires a more explicit treatment of recurrence, clustering, and return-time behavior than stress analysis usually receives. The framework developed here is meant to supply that treatment in a form that is mathematically legible and computationally reproducible.

Its contribution is therefore best described in bounded terms. The paper does not propose a new general theory of dependent extremes, nor does it resolve the empirical problems that arise when recurrence diagnostics are taken to real macro-financial data. What it does show is that recurrence can be formalized as a stress object, estimated through transparent pipeline components, summarized by an interpretable cluster-adjusted functional, and read alongside baseline volatility diagnostics without collapsing into them. The synthetic evidence supports that narrower claim, and the value of the paper lies in making the claim precisely enough that later empirical or theoretical work can either build on it or reject it on clear grounds.

References

[1] Tobias Adrian and Markus K. Brunnermeier. CoVaR. *American Economic Review*, 106(7):1705–1741, 2016.

[2] Viral V. Acharya, Lasse H. Pedersen, Thomas Philippon, and Matthew Richardson. Measuring systemic risk. *Review of Financial Studies*, 30(1):2–47, 2017.

[3] Dimitrios Bisias, Mark Flood, Andrew W. Lo, and Stavros Valavanis. A survey of systemic risk analytics. *Annual Review of Financial Economics*, 4:255–296, 2012.

[4] Tim Bollerslev. Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3):307–327, 1986.

[5] Christian Brownlees and Robert F. Engle. SRISK: A conditional capital shortfall measure of systemic risk. *Review of Financial Studies*, 30(1):48–79, 2017.

[6] Stuart Coles. *An Introduction to Statistical Modeling of Extreme Values*. Springer, London, 2001.

[7] Paul Embrechts, Claudia Klüppelberg, and Thomas Mikosch. *Modelling Extremal Events for Insurance and Finance*. Springer, Berlin, 1997.

[8] Robert F. Engle. Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. *Econometrica*, 50(4):987–1007, 1982.

[9] C. A. T. Ferro and Johan Segers. Inference for clusters of extreme values. *Journal of the Royal Statistical Society: Series B*, 65(2):545–556, 2003.

[10] Ana C. M. Freitas, Jorge M. Freitas, and Mike Todd. Hitting time statistics and extreme value theory. *Probability Theory and Related Fields*, 147:675–710, 2010.

[11] Masayuki Hirata, Benoît Saussol, and Sandro Vaienti. Statistics of return times: a general framework and new applications. *Communications in Mathematical Physics*, 206:33–55, 1999.

[12] T. Hsing, J. Hüsler, and M. R. Leadbetter. On the exceedance point process for a stationary sequence. *Probability Theory and Related Fields*, 78:97–112, 1988.

[13] M. R. Leadbetter, Georg Lindgren, and Holger Rootzén. *Extremes and Related Properties of Random Sequences and Processes*. Springer, New York, 1983.

[14] Valerio Lucarini, Davide Faranda, Ana C. M. Freitas, Jorge M. Freitas, Mark Holland, Tobias Kuna, Matteo Nicol, Mike Todd, and Sandro Vaienti. *Extremes and Recurrence in Dynamical Systems*. Wiley, Hoboken, 2016.

[15] George L. O'Brien. Extreme values for stationary and Markov sequences. *The Annals of Probability*, 15(1):281–291, 1987.

[16] Mária Süveges. Likelihood estimation of the extremal index. *Extremes*, 10:41–55, 2007.